

An Open-Hole Hénon Survivor and Its Exact Escape Determinant

Anonymous Route-A report

Abstract

We freeze the certified four-state local Hénon symbolic interface and remove one geometric state before every iterate. The surviving three-state matrix has exact determinant $1 - z - z^3$. We enumerate rooted survivor words, primitive necklaces, and escape traces through period twelve, with independent and symbolic checks. This is a discrete open-transfer certificate: it does not yet prove that a geometric open Hénon operator is analytic or nuclear, and it makes no arithmetic or Hilbert–Pólya claim.

1 Frozen hole

The inherited source/target adjacency is

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

We delete state 3 before each iterate. The survivor matrix is therefore

$$B = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

The hole is frozen; it is not selected after looking at traces or zeros.

2 Exact escape ledger

Direct expansion gives

$$\det(I - zB) = 1 - z - z^3.$$

If $t_n = \text{tr}(B^n)$ and p_n counts primitive survivor necklaces, then

$$t_n = \sum_{d|n} dp_d.$$

The first twelve rooted trace counts are

$$1, 1, 4, 5, 6, 10, 15, 21, 31, 46, 67, 98.$$

Proposition 1. *The open matrix, determinant, trace sequence, and primitive inversion are reproduced exactly by an independent implementation and a SymPy calculation.*

Proof. The checker reconstructs the principal submatrix from the frozen adjacency, computes matrix powers, and applies divisor inversion. The symbolic script independently expands the determinant and traces. $\square$

3 Controls and boundary

Deleting states 0, 1, 2, 3 instead gives determinants 1 , $1 - z$, $1 - z$, and $1 - z - z^3$, respectively. Thus the chosen hole is a genuine frozen geometry choice rather than a relabeling convention. The present theorem is only a finite symbolic open-transfer result. A future A2 upgrade must prove that the geometric hole has no boundary periodic orbits and that the same open transfer operator is compact/nuclear on a named space. The preregistered verdict is ‘A1_PARTIAL_CERTIFIED’, ‘A2_CERTIFIED_PREFIX’, ‘A3_NOT_ADDRESSED’, ‘A4_FAIL’. No prime table, zero table, Euler factor, or Route-B object is used.